

Tanvir Ahmed

✉ tanvirahmed123000@gmail.com | 📞 +8801906190296
🐙 github.com/tanvir-talha058 | 🔗 linkedin.com/in/tanvir-talha058

Skills

Languages: Python, C, C++, Java, JavaScript, Dart, PHP, SQL, PostgreSQL

AI / ML & Libraries: TensorFlow, PyTorch, scikit-learn, Keras, OpenCV, Mediapipe, NumPy, Pandas, Selenium, Tkinter, Matplotlib, Seaborn, LangChain, ChromaDB, Qdrant.

Web / Mobile / Tools: HTML, CSS, Bootstrap, Tailwind CSS, JavaScript, Flutter, Firebase, PHP, Git, GitHub, VS Code, Figma, OpenGL Google Colab, Kaggle, Tinkercad, Zapier, Make.

Concepts & Domains: Machine Learning, Deep Learning, Computer Vision, RAG, Chatbot, OCR, LLM, VLM Image/Video Processing, NLP, Object Tracking, IoT, Automation, Human-Computer Interaction, Data Visualization, Big Data, Information Retrieval.

Work Experience

AI/ML Engineer (Officer) - Strategy & Product
upay (UCB Fintech Company Limited)

2026 (June) - Present

- Designed an AI-powered outbound engagement platform featuring conversational AI, speech-to-text, LLM-based customer insight extraction, automated lead qualification, follow-ups, and campaign analytics.
- Designed an end-to-end cashless campus ecosystem for UCSI University, enabling digital payments across cafeterias, transportation, campus events, and student services while developing strategies for student onboarding, merchant acquisition, and QR payment adoption through upay.

AI/ML Intern - Digital Technology & Innovation (DTI)
United Commercial Bank PLC.

2026 (February) - 2026 (May)

- Built a production-grade multilingual RAG chatbot (English/Bangla/Banglish) for customer support using hybrid retrieval (dense embeddings + BM25 + reranking) and a fully local pipeline (scraping → preprocessing → embedding → vector indexing); implemented context control and query filtering to reduce hallucination and improve response accuracy.
- Developed a hybrid document understanding pipeline combining OCR, layout-aware region segmentation, template matching, and human-in-the-loop validation to improve handwritten text extraction accuracy, minimize manual processing overhead, and streamline banking workflow automation at scale.
- Developed a fraud detection system for real-time transaction monitoring with risk scoring based on velocity, frequency, and behavioral signals; designed a scalable architecture for future ML-based anomaly detection and customer behavioral profiling.

Student / Researcher (Freelance & Academic Projects)
Independent Projects & Research

2024–2025

- Built and implemented applied ML and automation projects during B.Sc.; collaborated with supervisors on data set curation, model design, and end-to-end apps.
- Designed and implemented diverse academic, personal, and client projects, applying practical ML and software engineering skills to real-world problems.

Education

Daffodil International University

2022–2025

B.Sc. in Computer Science & Engineering

Relevant Coursework: Object-Oriented Programming, Databases, Discrete Mathematics, Data Structures and Algorithms, Operating Systems, Computer Networks, Machine Learning and Data Mining, Image Processing, Computer Graphics, System Design, Software Engineering, Information Security, Statistics and Probability.

Cantonment College Jashore

2018–2020

Higher Secondary Certificate (HSC)

GPA: 5.00

Nabojibon Institute Satkhira

2016–2018

Secondary School Certificate (SSC)

GPA: 5.00

Project Work

CrimeMap BD <i>Python, Flask, PostgreSQL, JavaScript, Leaflet.js, Vercel, Supabase</i> - Developed a bilingual, responsive dashboard with geospatial mapping (Leaflet.js), dynamic filtering, and automated classification of 15+ Bangladeshi news sources across 64 districts using parallel HTTP fetching (32x speed improvement) and NLP-based deduplication - Built interactive geospatial dashboard with Leaflet.js clustering, heatmap, and Chart.js analytics; deployed serverless on Vercel + Supabase with sub-100ms API response times - Achieved 91 passing tests covering API routes, text classification pipelines, and event clustering logic	2024–2026
Automation Tool for Levi Strauss & Co. <i>Python, Selenium, Tkinter</i> Interactive app to upload Excel and auto-fill web forms with dynamic field mapping, reducing manual entry and errors.	2024–2025
AgroKart BD — Cross-Platform Marketplace <i>Flutter, Dart, Firebase, HTML/CSS/JS, PHP, SQL</i> - Developed a mobile app (Flutter) and a responsive web portal connecting farmers directly with consumers. - Implemented user authentication, product listings, cart and backend order management (PHP & SQL).	2024–2025
Edu-Buddy — AI Student Assistant <i>Python, SpeechRecognition, Google TTS, SQLite, Tkinter</i> - Voice-activated assistant for reminders, study schedules, and quick access to educational resources. - Modular intent parsing and dialog flow; integrated TTS and speech recognition for desktop use.	2023–2024
Hand Gesture Virtual Mouse <i>Python, OpenCV, Mediapipe, PyAutoGUI</i> - Real-time gesture-controlled cursor with click, scroll, and drag using multi-landmark detection. - Optimized frame pipeline for low-latency interaction on consumer hardware.	2023–2024
Cross-Platform Task Manager <i>Python, psutil, Tkinter</i> - Lightweight tool to monitor processes, CPU/memory usage, and perform safe terminations across platforms. - Boost and optimize the device for more than 10-12% of performance.	2023–2024

Research & Publications

Towards Automatic Classification and Translation of Bangla Regional Dialects in Low-Resource Settings <i>Natural Language Processing, Deep Learning, Transformer Models</i> Designed and developed a transformer-based NLP pipeline for Bangla regional dialect classification, leveraging a 20,090-sample dataset across 8 dialects to achieve 86	2024–Present
Truth in Pixel: Deepfake Detection <i>Computer Vision, Deep Learning</i> Working on pixel-level inconsistency detection and robust artifact-based pipelines for manipulation detection.	2024–Present
Exploiting Transformer Architectures for Domain-Specific Text Categorization: The m-BERT Perspective <i>Transformer Learning, Deep Learning, Natural Language Processing</i> Achieved benchmarking accuracy on this research among all existing works.	2024–2025
Plant Leaf Disease Recognition Dataset <i>Mendeley Data: Plant Leaf Disease Recognition Dataset</i> Curated and published 20000+ annotated plant leaf images (zucchini, hibiscus, gourd, papaya) to support CV research.	2023–2024
Deep Learning Models for Multi-Class Plant Leaf Disease Classification <i>CNN, Image Classification</i> Trained Multiple CNNs achieving validation accuracy exceeding 97% using augmentation and tuning.	2023–2024

Awards and Certificates

Selected Achievements - First Position — District Level Science Fair (Innovation Category) - Third Position — Division Level Science Fair (Innovation Category)
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